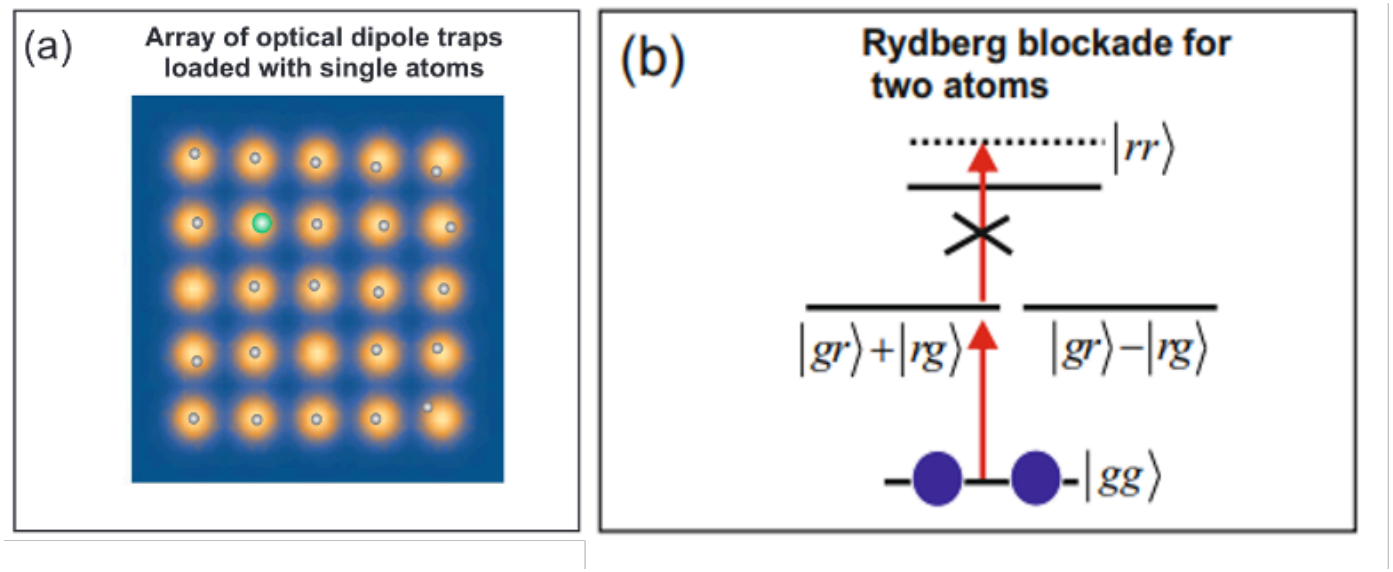
Main contents

- Difficulties of quantum gate construction in neutral-atom system.
- Adiabatic passage: Adiabatic rapid passage.
- Quantum gate construction with ARP pulse.

Quantum computing in neutral-atom system

- High-fidelity two qubit gate based on blockade effect.[1]

– Difficulty: single-atom loss.

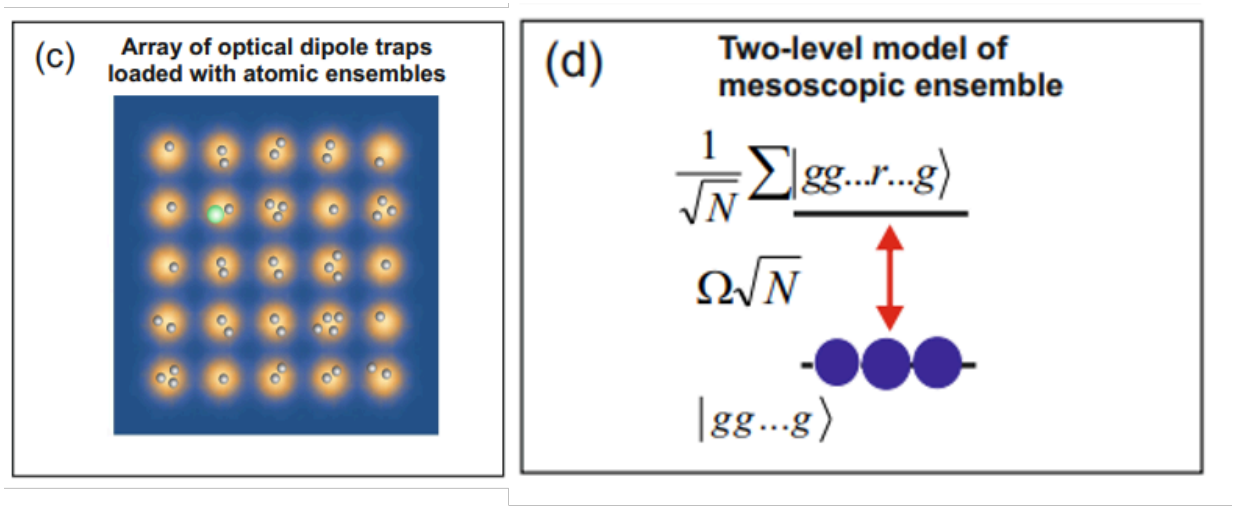


- Mesoscopic atomic ensemble with single excitation can be regarded as a qubit.[1]

Quantum computing in neutral-atom system

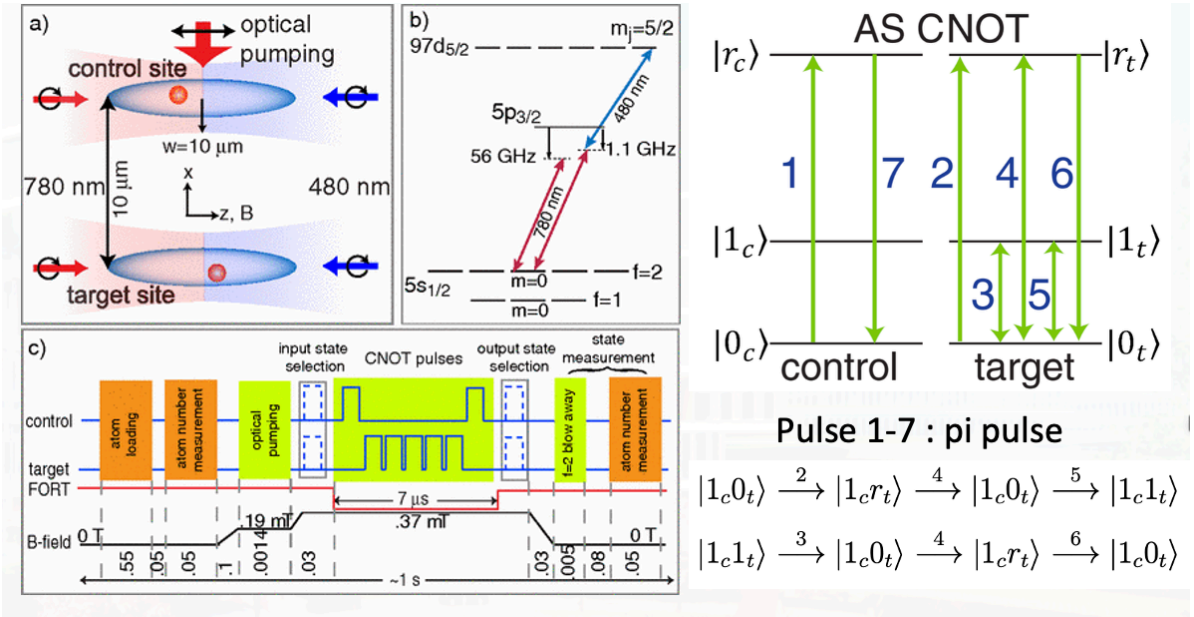
– advantage: reduced sensitivity to atom losses; enhanced coupling to light.

– difficulty: fluctuation of the number of atoms in the ensemble.



Traditional π-pulse methods

- Highly sensitive to variations in pulse area.



$$|1_c 0_t\rangle \xrightarrow{2} |1_c r_t\rangle \xrightarrow{4} |1_c 0_t\rangle \xrightarrow{5} |1_c 1_t\rangle$$

$$|1_c 1_t\rangle \xrightarrow{3} |1_c 0_t\rangle \xrightarrow{4} |1_c r_t\rangle \xrightarrow{6} |1_c 0_t\rangle$$

ensemble with unknown number of atoms

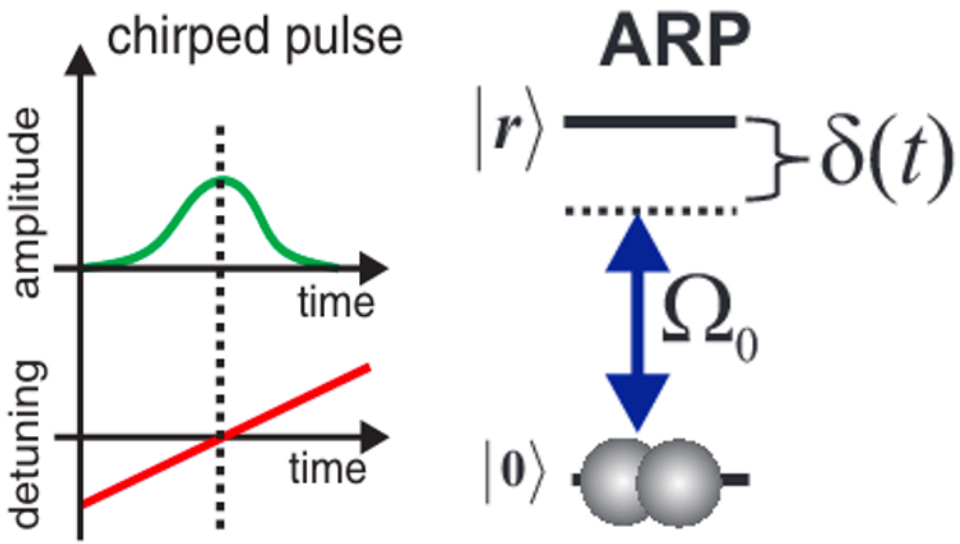
- ensemble: pulse area = $\sqrt{N} * \Omega * t = \pi$.
- randomly loaded dipole trap follows a Poissonian distribution in atom number. (unknown N)

Adiabatic passage

- Two-level system transfer: Adiabatic rapid passage(ARP)
- Three-level system: Stimulated Raman adiabatic passage(STIRAP).

Adiabatic rapid passage(ARP)

- Two-level system transfer.[1]
- Driven with chirped pulse.[2]



Why ARP?

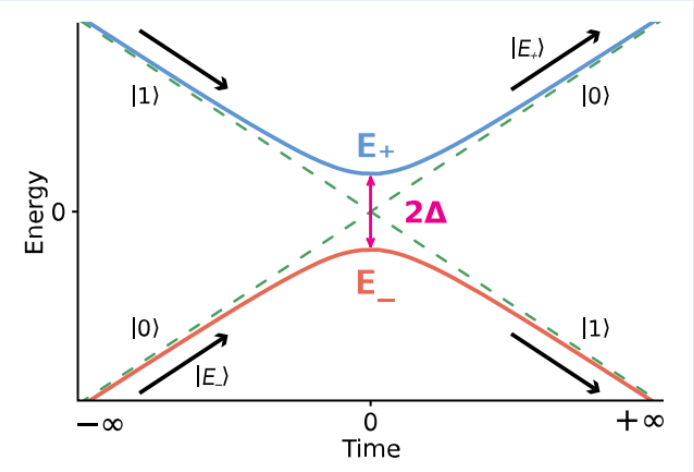
- Point 1: 100% transition probability under adiabatic condition.

ARP and LZ theory

Landau-Zener theory[3], [4]

- Linear sweep of the energy level.

$$H = \frac{\mathrm{vt}}{2}\sigma_z + \frac{\Omega}{2}\sigma_x$$



ARP and LZ theory

- Transition probability:

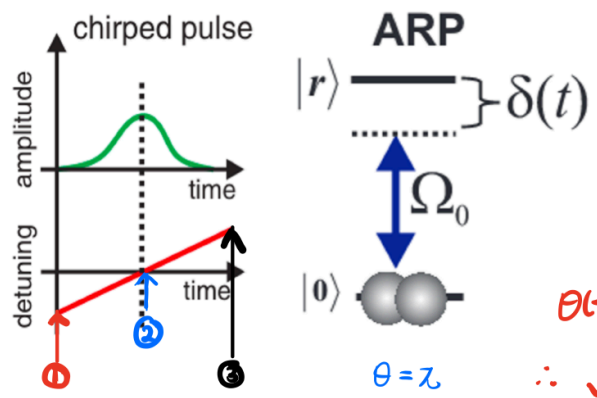
$$P = \exp\left(-\frac{\pi\Omega^2}{2v}\right)$$

if $v \ll \Omega^2$(Slow varying system, $P = 0$): adiabatic transition from $|0\rangle$ to $|1\rangle$ during $t = -\infty$ to $t = \infty$.

Connection between ARP and LZ

- Two-level system:[1], [2]

adiabatically following the eigenstate.



$$H = \begin{pmatrix} 0 & \Omega(t)/2 \\ \Omega(t)/2 & \Delta(t) \end{pmatrix}$$

$\downarrow$ diagonalise $\rightarrow U^{+} H U$

$$U(t) = \begin{pmatrix} \cos\frac{\theta(t)}{2} & -\sin\frac{\theta(t)}{2} \\ \sin\frac{\theta(t)}{2} & \cos\frac{\theta(t)}{2} \end{pmatrix}$$

$$\theta(t) = \arctan\left(\frac{\Omega(t)}{\Delta(t)}\right)$$

$$\therefore \quad |\psi_+(t)\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$$

$$|\psi_-(t)\rangle = \sin\frac{\theta}{2}|0\rangle + \cos\frac{\theta}{2}|1\rangle$$

① : $\frac{\Omega}{\Delta} \rightarrow 0$; $\frac{\Omega}{\Delta} < 0$ ② $\frac{\Omega}{\Delta}$: $-\infty \rightarrow +\infty$ ③ $\frac{\Omega}{\Delta} \rightarrow 0$, $\frac{\Omega}{\Delta} > 0$ $\theta = 0$

$\theta = \pi$ $|\psi_+\rangle = |1\rangle$ $\theta = \frac{\pi}{2}$ $|\psi_+\rangle = |+\rangle$ $|\psi_+\rangle = |2\rangle$

Why ARP?

- Point 2:

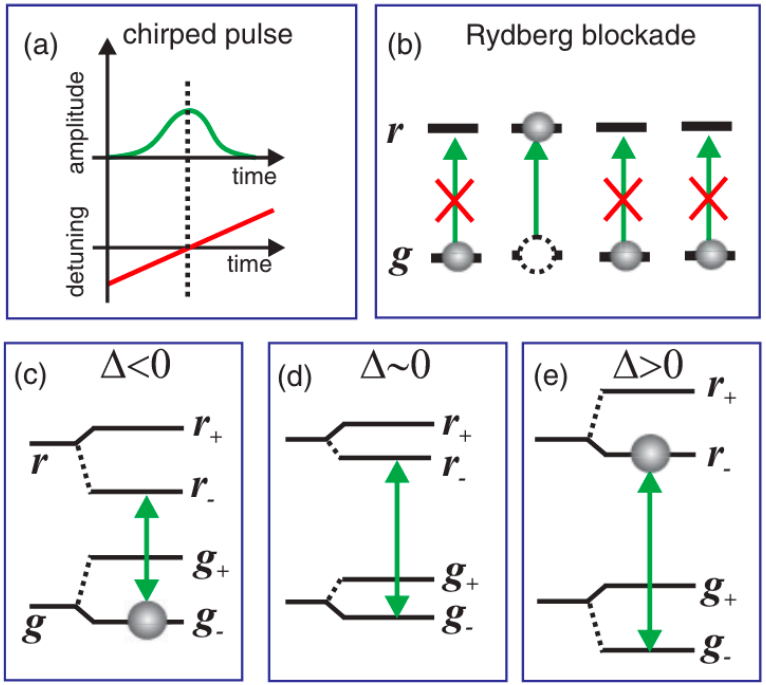
– Insitive to pulse areas[2], [5].

– Overcoming the problem of the $\sqrt{N}$ dependence of the collective Rabi frequency in the ensemble with unknown number of atoms.[2]

– As mentioned above, the Rabi frequency is only required to meet the adiabatic condition and the pulse area does not influence the transition probability.

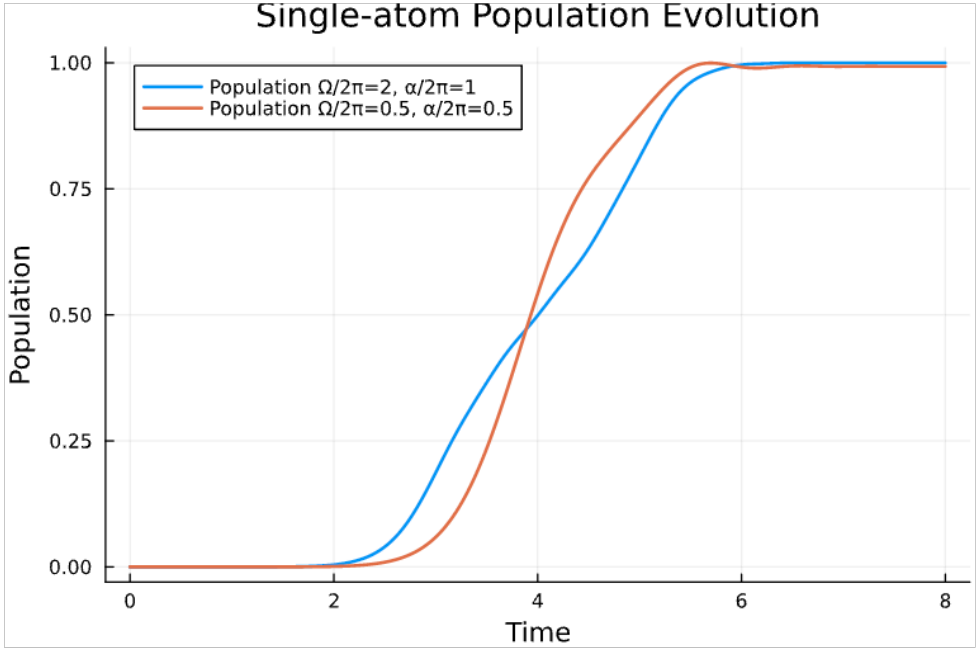
Deterministic single-excitation with ARP

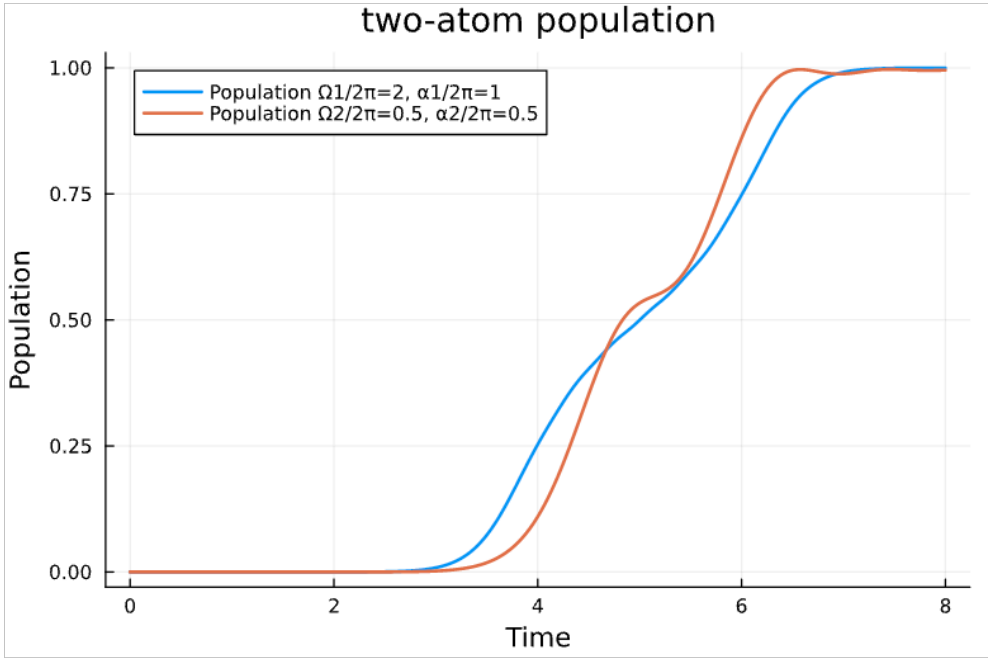
- Linear chirped Gaussian pulse with blockade condition.[2]



- Time-dependence of excitation probability with different chirped rate:

Deterministic single-excitation with ARP





- The probability of single-atom Rydberg excitation is independent of the number of atoms.

Why ARP?

- Point3:

– The phase accumulation is not affected by N.[1]

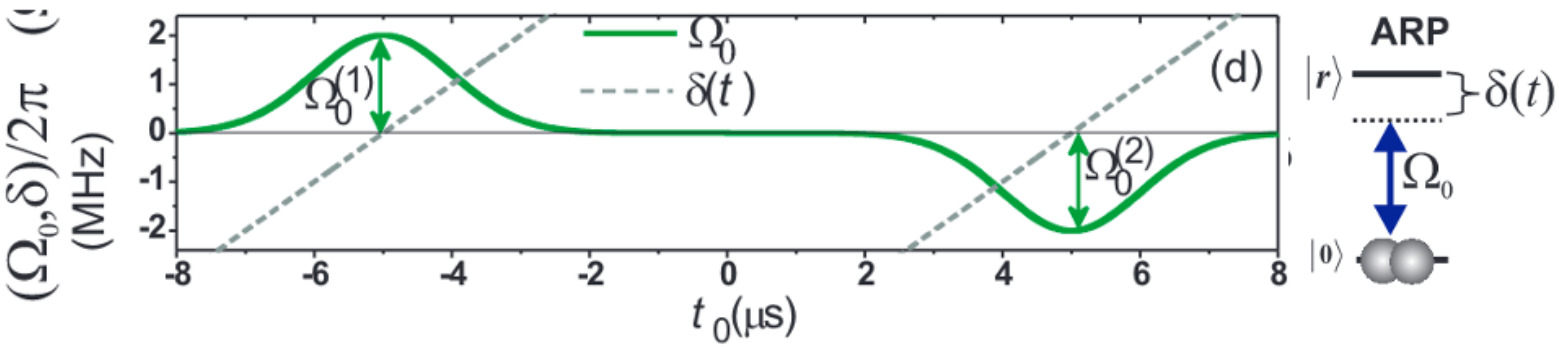
– performing high-fidelity quantum gate without precise knowledge of the ensemble number.

Two-level system with ARP pulse

$$H(t) = \begin{pmatrix} 0 & \Omega(t)/2 \\ \Omega(t)/2 & -\Delta(t) \end{pmatrix}$$

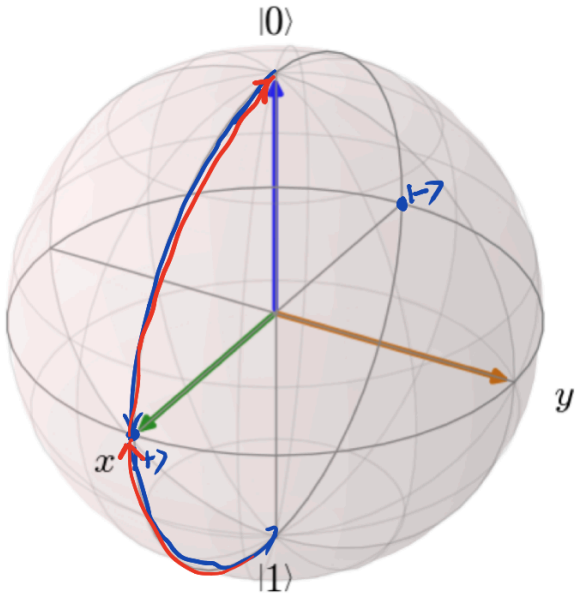
$$\Delta(t) = \alpha * t; \Omega(t) = \exp\left(-\frac{t^2}{2\tau}\right)$$

where $\alpha/2\pi = 1\text{MHz}; \Omega/2\pi = 2\text{MHz}, \tau = 1\mu s$



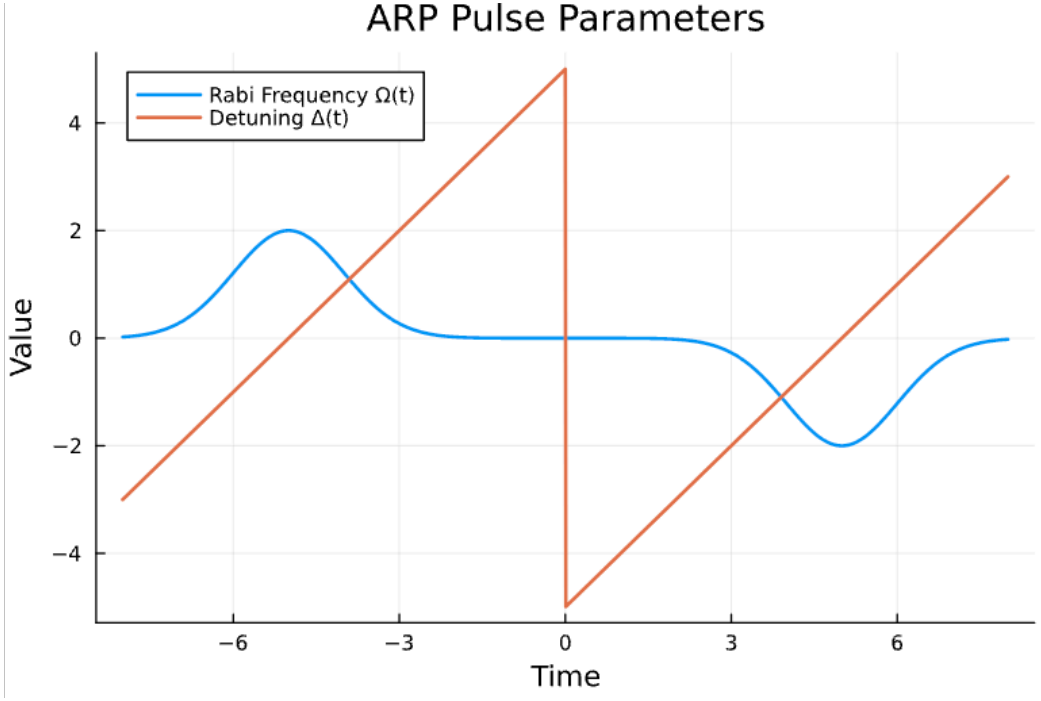
Two-level system with ARP pulse

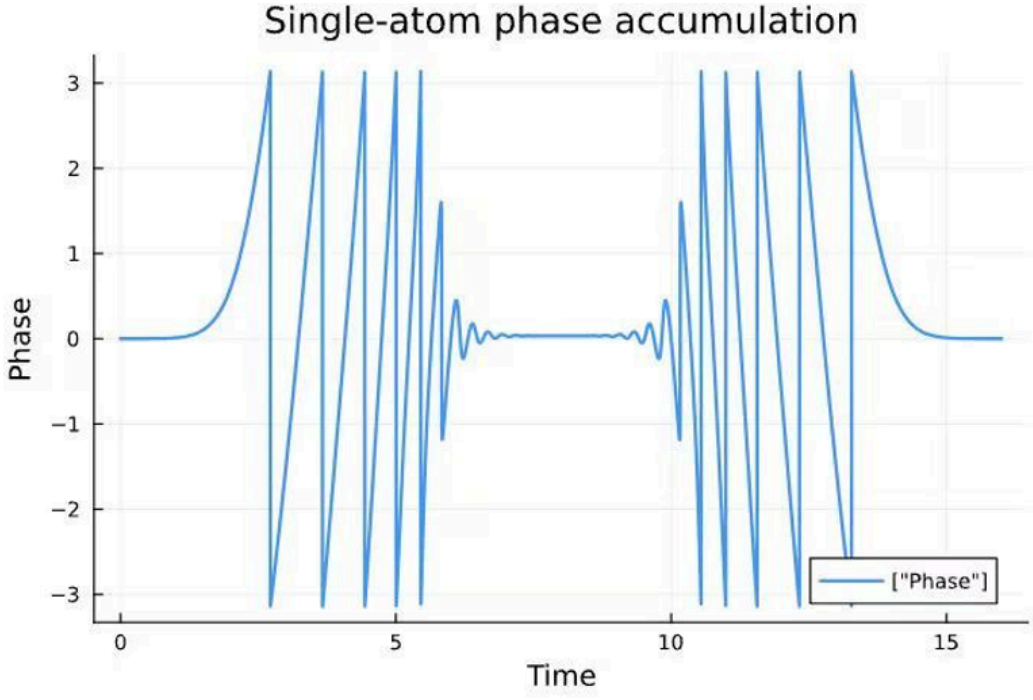
- phase accumulation is zero
- state transition: $|0\rangle \to |+\rangle \to |r\rangle \to |+\rangle \to |0\rangle$



Two-level system with chirped pulse

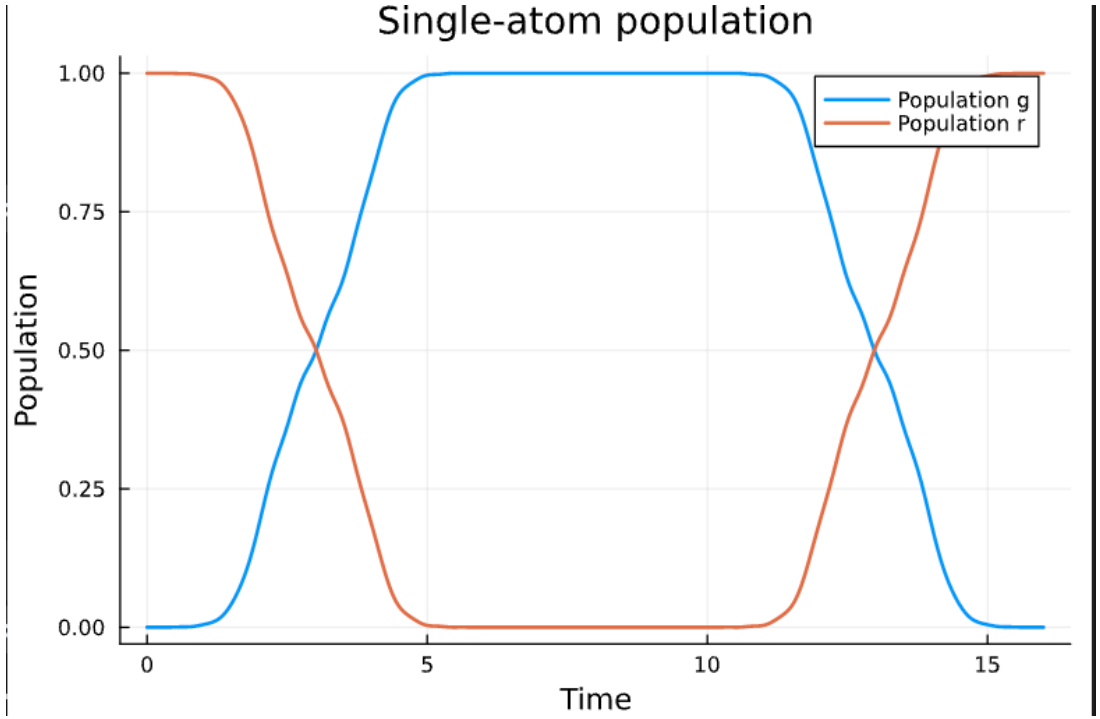
- for one single excitation, the phase accumulation is





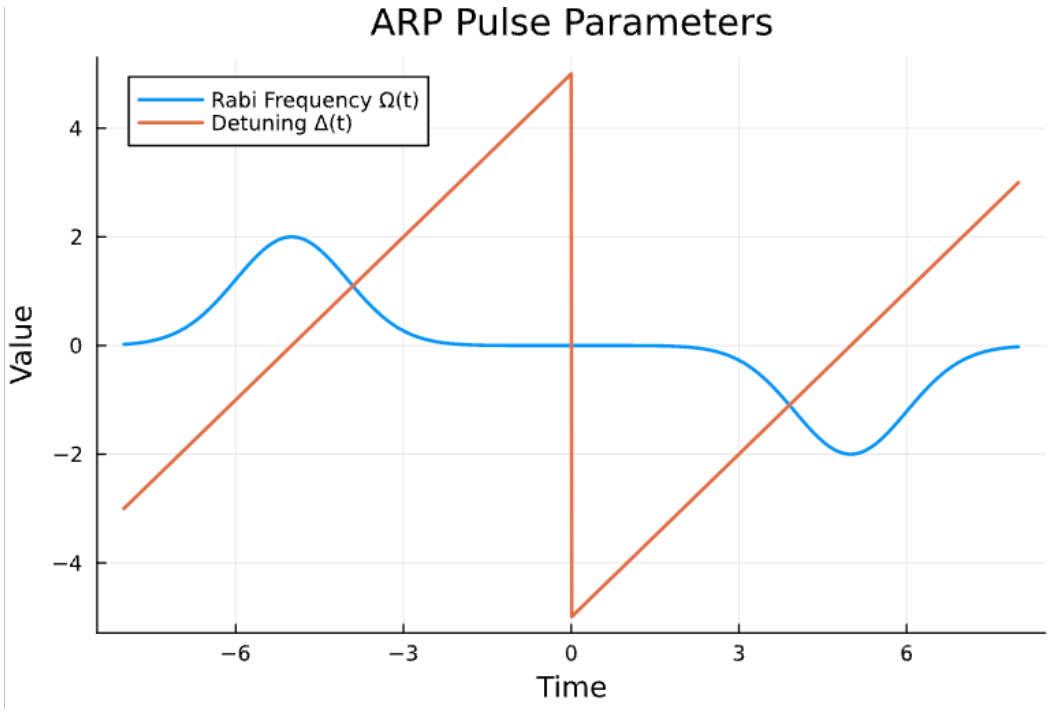
- The corresponding population is:

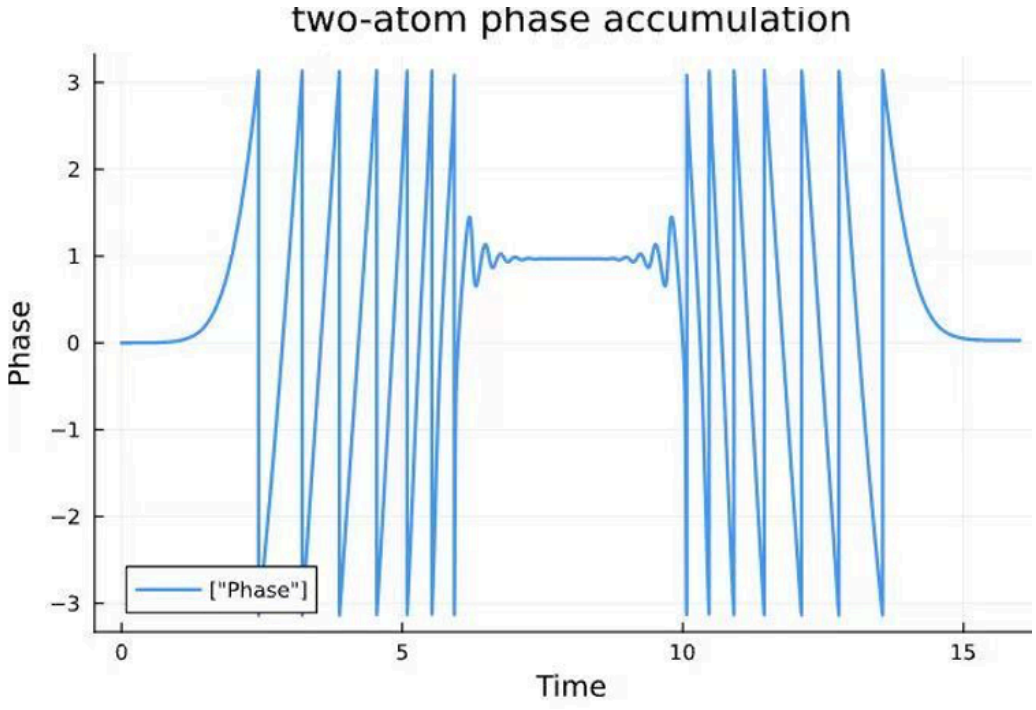
Two-level system with chirped pulse



Two-level system with chirped pulse

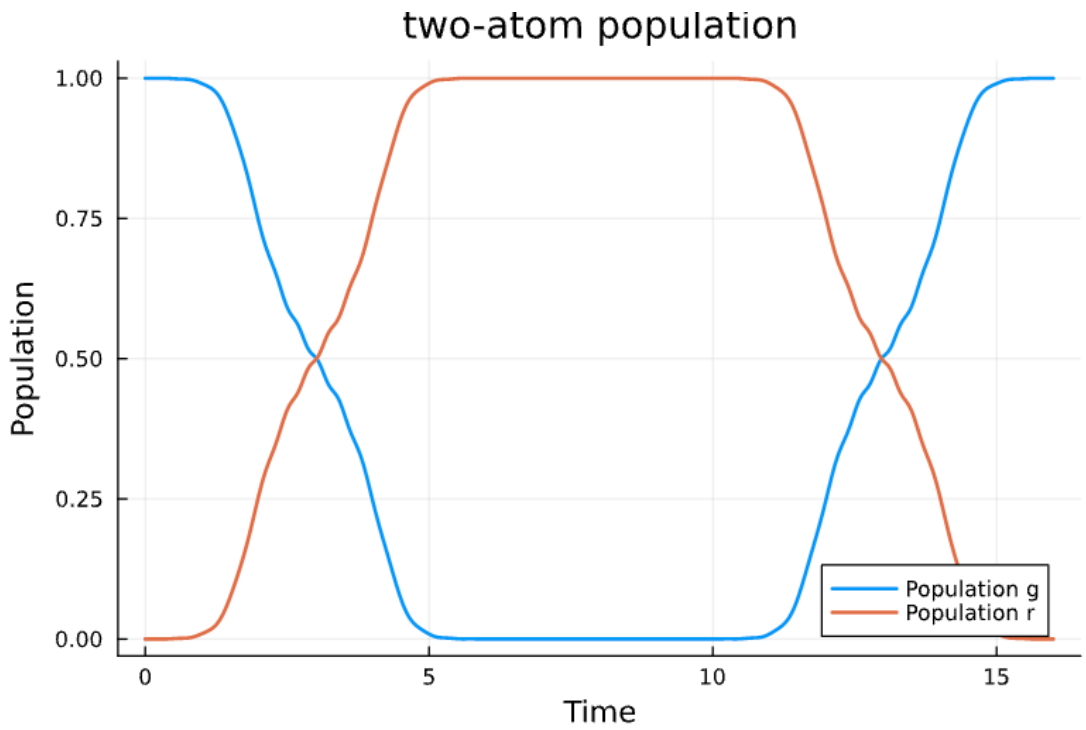
- for two single excitation, the phase accumulation is





- The corresponding population is:

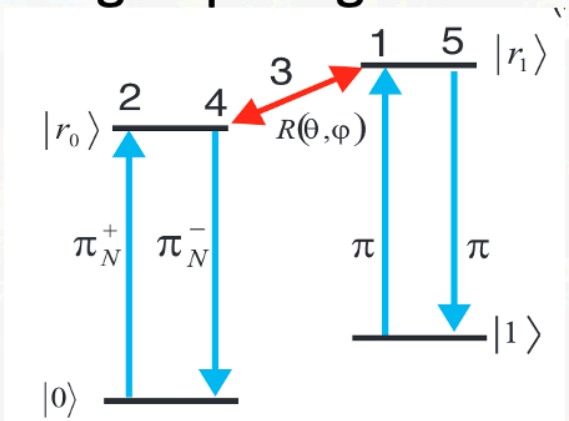
Two-level system with chirped pulse



Gate design with ARP pulse

Single qubit gate:

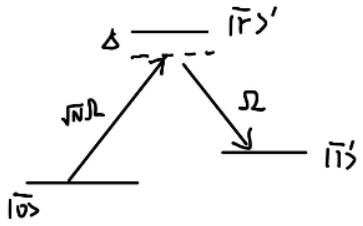
$$|\psi\rangle = a\,|\bar{0}\rangle + b\,|\bar{1}\rangle$$



$$|\psi_1\rangle = a\,|\bar{0}\rangle + ib\,|\bar{r}_1\rangle$$
$$|\psi_2\rangle = a\,|\bar{r}_0\rangle + ib\,|\bar{r}_1\rangle$$
$$|\psi_3\rangle = a'\,|\bar{r}_0\rangle - ib'\,|\bar{r}_1\rangle$$
$$|\psi_4\rangle = a'\,|\bar{0}\rangle - ib'\,|\bar{r}_1\rangle$$
$$|\psi_5\rangle = a'\,|\bar{0}\rangle + b'\,|\bar{1}\rangle\,.$$

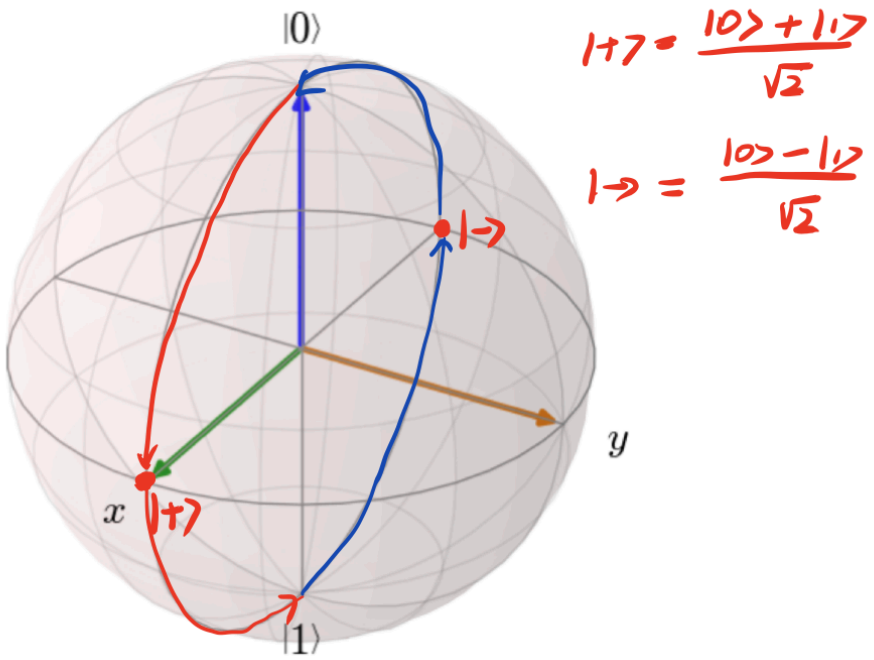
$$|0\rangle = |00\cdots0\rangle$$
$$|\bar{1}\rangle' = \frac{1}{\sqrt{N}}\sum_j |00\ldots1_j\ldots00\rangle$$
$$|\bar{r}\rangle' = \frac{1}{\sqrt{N}}\sum_j |00\ldots r_j\ldots00\rangle$$
$$|\bar{1}\rangle = e^{i\chi_N}|\bar{1}\rangle'$$
$$|\bar{r}\rangle = e^{i\chi_N}|\bar{r}\rangle'$$

STIRAP pulse phase



Gate design with ARP pulse

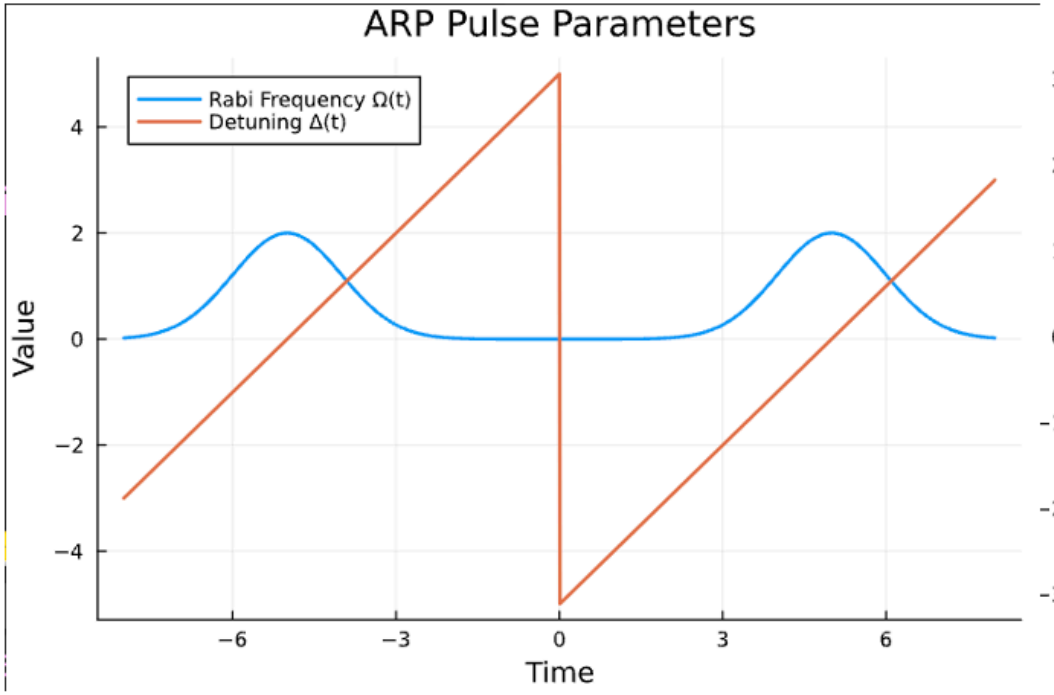
- if we do not switch the phase:

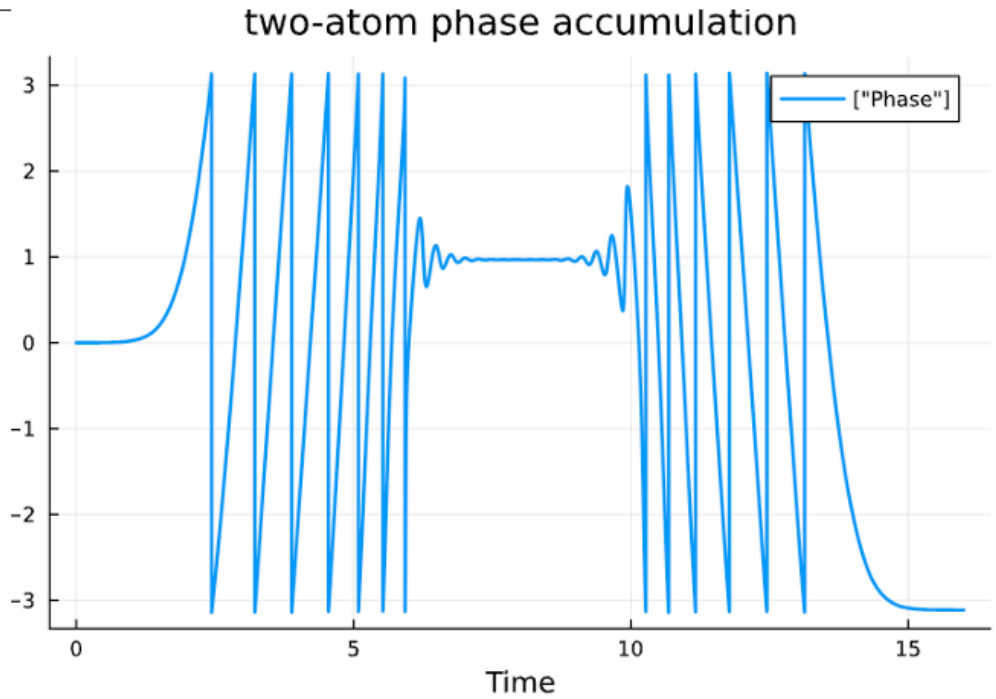


$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

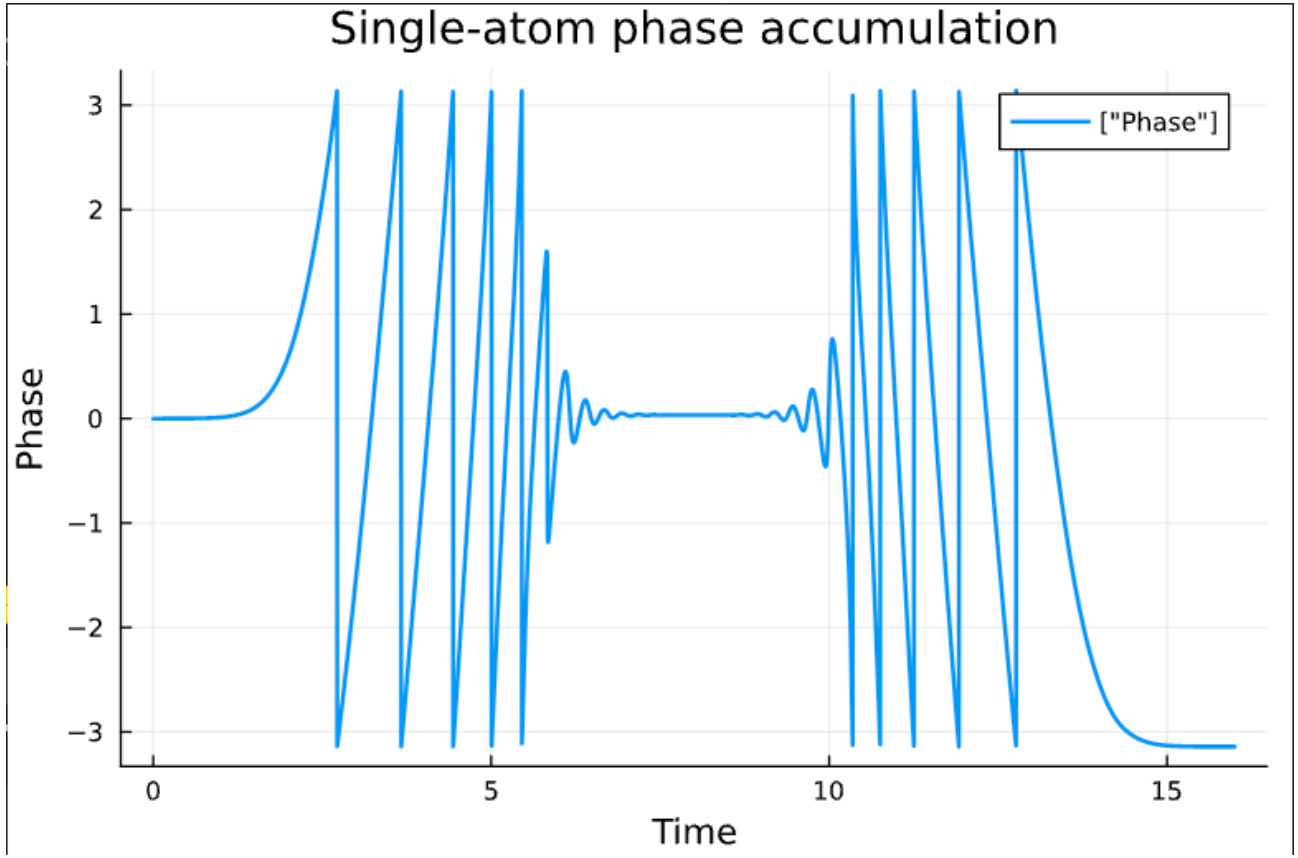
$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Gate design with ARP pulse

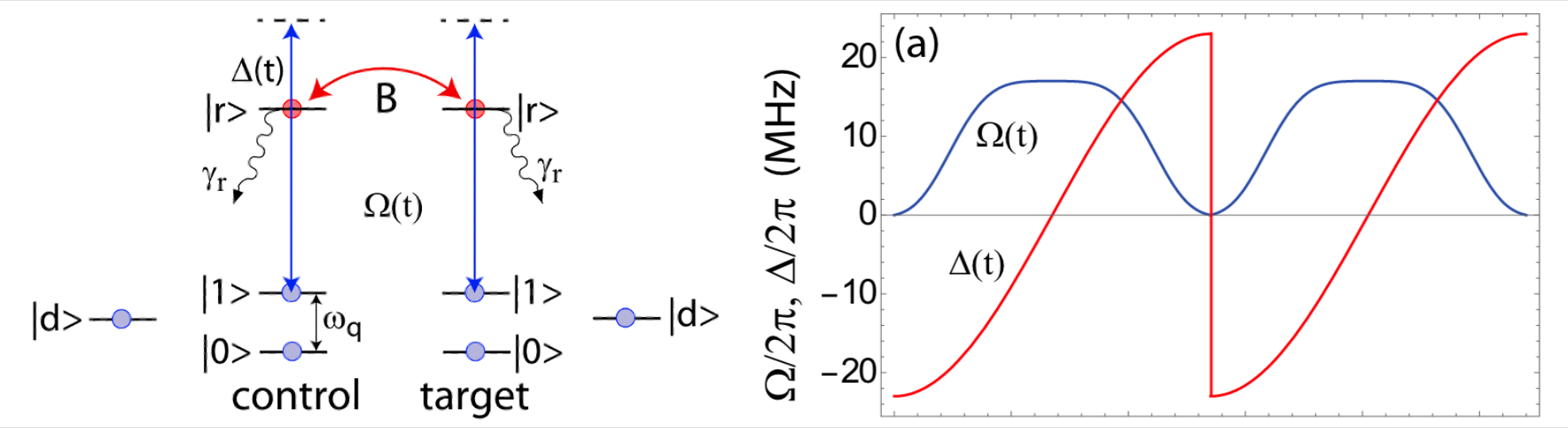




Gate design with ARP pulse



Gate design with ARP pulse[6]



- Basis: $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$

– Prepare in state $|00\rangle$: We neglect optical excitation of the $|0\rangle$ state due to the large detuning $\omega_q \gg |\Omega|$

– Prepare in state $|10\rangle$ or $|10\rangle$:

Gate design with ARP pulse[6]

$$H(t) = \begin{pmatrix} 0 & \Omega(t)/2 \\ \Omega(t)/2 & -\Delta(t) \end{pmatrix}$$

– Prepare in state $|11\rangle$:

$$H(t) = \begin{pmatrix} 0 & \sqrt{2}\Omega(t)/2 \\ \sqrt{2}\Omega(t)/2 & -\Delta(t) \end{pmatrix}$$

The Rabi frequency in ARP pulse is only determined for satisfying adiabatic condition.

- As mentioned above, the phase accumulation of both state $|10\rangle$, $|01\rangle$ and $|11\rangle$ is π.

Application of ARP

- Useful technique for state preparation in quantum dots[7], population inversion[8] and quantum computing.
- Frequency-swept picosecond laser pulses can be used to achieve efficient population transfer by ARP in two-level and multi-level systems in the adiabatic limit.[9]
- A general theory of ARP with intense, linearly chirped laser pulses with a modified LZ formula to determine the optimal conditions for efficient and robust population transfer. [5]
- Coherent population transfer between Rydberg states of Li atoms with ARP of a single multiphoton transition.alert([10]

ARP process considering relaxing

- Considerign the effect of dissipation induced by environmental coupling.[11], [12], [13]. They showed a non-monotonic dependence of the transition probability on the sweep speed due to competition between relaxation and the external sweep. [14], [15], [16], [17]
- Also, strong drive can also result in driven induced dessipation(DID), which deserves further investigation.[18]

development of ARP

- Initial experiments: fixed laser frequency and sweeped DC field.[19], [20]
- More recent methods: STIRAP(Stimulated Raman Adiabatic Passage), APLIP(Adiabatic Passage with Laser Induced Potentials).

Chirped pulse in ARP

- Sweeping laser frequency with a linear chirpe.

$$\omega(t) = \omega_0 + \alpha t$$

$$E(t) = E_0 \exp\left(-\frac{t^2}{2\tau} - i\omega_0 t - i\alpha\frac{t^2}{2}\right)$$

Thus, the hamiltonian of two-level system with chirped pulse is:

$$H = \frac{\Delta(t)}{2}\sigma_z + \frac{\Omega}{2}\sigma_x$$

Adiabatic condition

- Adiabatic condition:

$$\frac{\left|\langle\psi_+|\ \frac{\mathrm{dH}}{\mathrm{dt}}\ |\psi_-\rangle\right|}{\left(E_+ - E_-\right)^2} \ll 1$$

- Under two-level system($H = \frac{\mathrm{vt}}{2}\sigma_z + \frac{\Omega}{2}\sigma_x$):

$$\frac{v\Omega}{2(\Delta^2 + \Omega^2)^{\frac{3}{2}}} \ll 1$$

when $\Delta = 0$, $v \ll 2\Omega^2$, we need to sweep slow enough.

Bibliography

Bibliography

[1] I. I. Beterov *et al.*, "Application of adiabatic passage in Rydberg atomic ensembles for quantum information processing," *Journal of Physics B: Atomic, Molecular and Optical Physics*, vol. 53, no. 18, p. 182001, Jul. 2020, doi: 10.1088/1361-6455/ab8719.

[2] I. I. Beterov *et al.*, "Deterministic single-atom excitation via adiabatic passage and Rydberg blockade," *Phys. Rev. A*, vol. 84, no. 2, p. 23413, Aug. 2011, doi: 10.1103/PhysRevA.84.023413.

[3] J. R. Rubbmark, M. M. Kash, M. G. Littman, and D. Kleppner, "Dynamical effects at avoided level crossings: A study of the Landau-Zener effect using Rydberg atoms," *Phys. Rev. A*, vol. 23, no. 6, pp. 3107–3117, Jun. 1981, doi: 10.1103/PhysRevA.23.3107.

Bibliography

[4] F. Kahlert, V. Link, R. Hartmann, and W. T. Strunz, "Simulating the Landau–Zener sweep in deeply sub-Ohmic environments," *The Journal of Chemical Physics*, vol. 161, no. 18, p. 184108, 2024, doi: 10.1063/5.0235741.

[5] V. S. Malinovsky and J. L. Krause, "General theory of population transfer by adiabatic rapid passage with intense, chirped laser pulses," *The European Physical Journal D - Atomic, Molecular, Optical and Plasma Physics*, vol. 14, no. 2, pp. 147–155, doi: 10.1007/s100530170212.

[6] M. Saffman, I. I. Beterov, A. Dalal, E. J. Páez, and B. C. Sanders, "Symmetric Rydberg controlled-Z gates with adiabatic pulses", *Phys. Rev. A*, vol. 101, no. 6, p. 62309, Jun. 2020, doi: 10.1103/PhysRevA.101.062309.

Bibliography

[7] A. Mukherjee *et al.*, "Electrically controlled rapid adiabatic passage in a single quantum dot," *Applied Physics Letters*, vol. 116, no. 25, p. 251103, 2020, doi: 10.1063/5.0012257.

[8] J. Feilhauer *et al.*, "Encircling exceptional points as a non-Hermitian extension of rapid adiabatic passage," *Phys. Rev. A*, vol. 102, no. 4, p. 40201, Oct. 2020, doi: 10.1103/PhysRevA.102.040201.

[9] J. S. Melinger, S. R. Gandhi, A. Hariharan, D. Goswami, and W. S. Warren, "Adiabatic population transfer with frequency-swept laser pulses," *The Journal of Chemical Physics*, vol. 101, no. 8, pp. 6439–6454, 1994, doi: 10.1063/1.468368.

[10] H. Maeda, J. H. Gurian, D. V. L. Norum, and T. F. Gallagher, "Coherent Population Transfer in an Atom by Multiphoton Adiabatic Rapid Passage," *Phys. Rev. Lett.*, vol. 96, no. 7, p. 73002, Feb. 2006, doi: 10.1103/PhysRevLett.96.073002.

Bibliography

[11] P. Ao and J. Rammer, "Quantum dynamics of a two-state system in a dissipative environment," *Phys. Rev. B*, vol. 43, no. 7, pp. 5397–5418, Mar. 1991, doi: 10.1103/PhysRevB.43.5397.

[12] P. Ao and J. Rammer, "Influence of dissipation on the Landau-Zener transition," *Phys. Rev. Lett.*, vol. 62, no. 25, pp. 3004–3007, Jun. 1989, doi: 10.1103/PhysRevLett.62.3004.

[13] P. Nalbach and M. Thorwart, "Landau-Zener Transitions in a Dissipative Environment: Numerically Exact Results," *Phys. Rev. Lett.*, vol. 103, no. 22, p. 220401, Nov. 2009, doi: 10.1103/PhysRevLett.103.220401.

[14] M. Wubs, K. Saito, S. Kohler, P. Hänggi, and Y. Kayanuma, "Gauging a Quantum Heat Bath with Dissipative Landau-Zener Transitions," *Phys. Rev. Lett.*, vol. 97, no. 20, p. 200404, Nov. 2006, doi: 10.1103/PhysRevLett.97.200404.

Bibliography

[15] Z. Sun, L. Zhou, G. Xiao, D. Poletti, and J. Gong, "Finite-time Landau-Zener processes and counterdiabatic driving in open systems: Beyond Born, Markov, and rotating-wave approximations," *Phys. Rev. A*, vol. 93, no. 1, p. 12121, Jan. 2016, doi: 10.1103/PhysRevA.93.012121.

[16] R. S. Whitney, M. Clusel, and T. Ziman, "Temperature Can Enhance Coherent Oscillations at a Landau-Zener Transition," *Phys. Rev. Lett.*, vol. 107, no. 21, p. 210402, Nov. 2011, doi: 10.1103/PhysRevLett.107.210402.

[17] S. Ashhab, "Landau-Zener transitions in an open multilevel quantum system," *Phys. Rev. A*, vol. 94, no. 4, p. 42109, Oct. 2016, doi: 10.1103/PhysRevA.94.042109.

[18] N. Chanda, P. Patnaik, and R. Bhattacharyya, "Optimal population transfer using adiabatic rapid passage in the presence of drive-induced dissipation," *Physical Review A*, vol. 107, no. 6, Jun. 2023, doi: 10.1103/physreva.107.063708.

Bibliography

[19] M. M. T. Loy, "Two-Photon Adiabatic Inversion," *Phys. Rev. Lett.*, vol. 41, no. 7, pp. 473–476, Aug. 1978, doi: 10.1103/PhysRevLett.41.473.

[20] M. M. T. Loy, "Observation of Population Inversion by Optical Adiabatic Rapid Passage," *Phys. Rev. Lett.*, vol. 32, no. 15, pp. 814–817, Apr. 1974, doi: 10.1103/PhysRevLett.32.814.